

Vrishab Prasanth Davey

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EDUCATION

University of Ottawa

Ottawa, Ontario

MCS and Concentration Applied Artificial Intelligence

- Computer Vision, NLP, AI Testing, BioInformatics

SRM Institute of Science and Technology

Chennai, India

BTech in Computer Science and Engineering(GPA: 3.8)

- Excellence awardee, Overall 2nd place in FB'22, AIR 50 - NEO, Research Assistantship

WORK EXPERIENCE

SRM Institute of Science & Technology

Chennai, India

Research Intern

Dec 2023 - May 2024

- **Developed** a lightweight U-Net model for **brain tumor segmentation** on the **BITE MRI dataset**, achieving **84% mean IoU** across **846 images**, surpassing traditional methods without data augmentation for faster, reliable clinical use.
- **Enhanced model robustness** by training on **coronal, sagittal, and transversal MRI planes**, increasing mean IoU by **5%** across views, **with 99%** pixel accuracy on the transversal plane, ensuring **multi-perspective diagnostic reliability**.
- **Benchmarked** U-Net against **K-Means, Fuzzy C-Means, and LinkNet**, achieving a **4% higher mean IoU (up to 89%)** and **greater pixel accuracy**, highlighting its real-time clinical segmentation capabilities.

Wipro

Chennai, India

Intern

Jan 2024 - Mar 2024

- **Designed a Virtual Agent** in ServiceNow, **activating the Glide Virtual Agent plugin** and **integrating ITSM Virtual Agent** from the ServiceNow Store, improving **ITSM response times by 30%**.
- **Configured 20+ conversation flows** in **Virtual Agent Designer** and **optimized Natural Language Understanding (NLU)** for **95% accuracy** in intent recognition.
- **Integrated OpenAI API** to handle complex queries, **fine-tuned a custom Retrieval-Augmented Generation (RAG) model**, achieving **90%+ accuracy in document-based responses** and significantly **reducing manual lookup time**.
- **Deployed as a Flask application**, providing user-friendly access, with adoption across multiple teams and a **40% increase in self-service resolution rates**.

Samsung R&D Institute India

Bangalore, India

Research Intern

Mar 2023 - Nov 2023

- **Spearheaded the development** of a **warping-based CNN for real-time Eye Gaze Correction** on live video, achieving **33 FPS** on consumer laptops, **addressing parallax issues**, and improving eye contact in video calls.
- **Enhanced gaze correction accuracy** by implementing **shape- and color-based loss functions**, reducing **sclera and eyelid artifacts by 20%**, and increasing **PSNR by 4 dB** over models like DeepWarp.
- **Validated model on diverse datasets**, including **DIRL and CAVE**, achieving **84.5% realism recognition** across ethnicities and **consistent gaze correction** within a **±30-degree range**.
- **Optimized model with architectural refinements** in TensorFlow, reducing **computation time by 20%** and enabling **seamless gaze correction in live video streaming**.

PROJECTS

1. U-Net for MRI Brain Tumor Segmentation

Developed a U-Net model for automated MRI brain tumor segmentation, **achieving 93.86% accuracy** and a **high Dice coefficient**. Integrated a Streamlit interface for real-time visualization, enhancing accessibility in **neuro-oncology workflows**.

2. Eye Gaze Correction for Video Conferencing

Constructed a real-time Eye Gaze Correction system with a **warping-based CNN**, achieving **33 FPS** and reducing eye contact issues by **minimizing parallax**. Custom **loss functions reduced visual artifacts by 20%**, validated across datasets with an **84.5% realism recognition rate**.

3. Blood Glucose Prediction using RNNs

Developed an RNN model to predict short-term blood glucose levels for **type 1 diabetes patients**, achieving **31.4 RMSE** for **60-minute forecasts**. Used **Gaussian distribution modeling** for uncertainty estimates and a customized **error grid** for clinical safety in closed-loop insulin systems.

SKILLS & INTERESTS

Programming Languages: Python, SQL, PostgreSQL, C, C++, TensorFlow, Keras, scikit-learn, Pandas, Numpy, Matplotlib, Seaborn, SQL, MLOps, NLTK, AWS, Flask, BeautifulSoup, Nodejs, CSS, Git, Docker

Certifications: Machine Learning Specialization, Introduction to Data Science Generative, AWS Academy Cloud Foundations, Oracle Database Foundations Certified Junior Associate, Database Management Essentials.

Publications: 1. Vrishab Davey, Dr. Akilandeswari P. "Streamlined U-Net for MRI Brain Tumor Segmentation with Streamlit Visualization Interface." IC2E3, IEEE, 2024. 2. Vrishab Davey, Dr. Akilandeswari P. "Eye Contact During Live Video Conferencing." IC2E3, IEEE, 2024.